

Lab Report: AWS Cloud Lab - Serverless vs Containers

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Lab: 4

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Assignment

Compare **AWS Lambda** (Zip & Container), **ECS Fargate**, and **EC2** using a k-NN search workload (50,000 vectors).

1. **Deploy** all four environments and verify identical results.
 2. **Scenario A:** Measure and decompose cold start latency.
 3. **Scenario B:** Measure warm steady-state throughput at varying concurrency.
 4. **Scenario C:** Analyze performance during a "burst from zero" traffic spike.
 5. **Cost Analysis:** Calculate idle costs and build a monthly cost model.
 6. **Recommendation:** Provide a quantified infrastructure choice based on a p99 < 500ms SLO.
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Solution

1. Deployment Verification

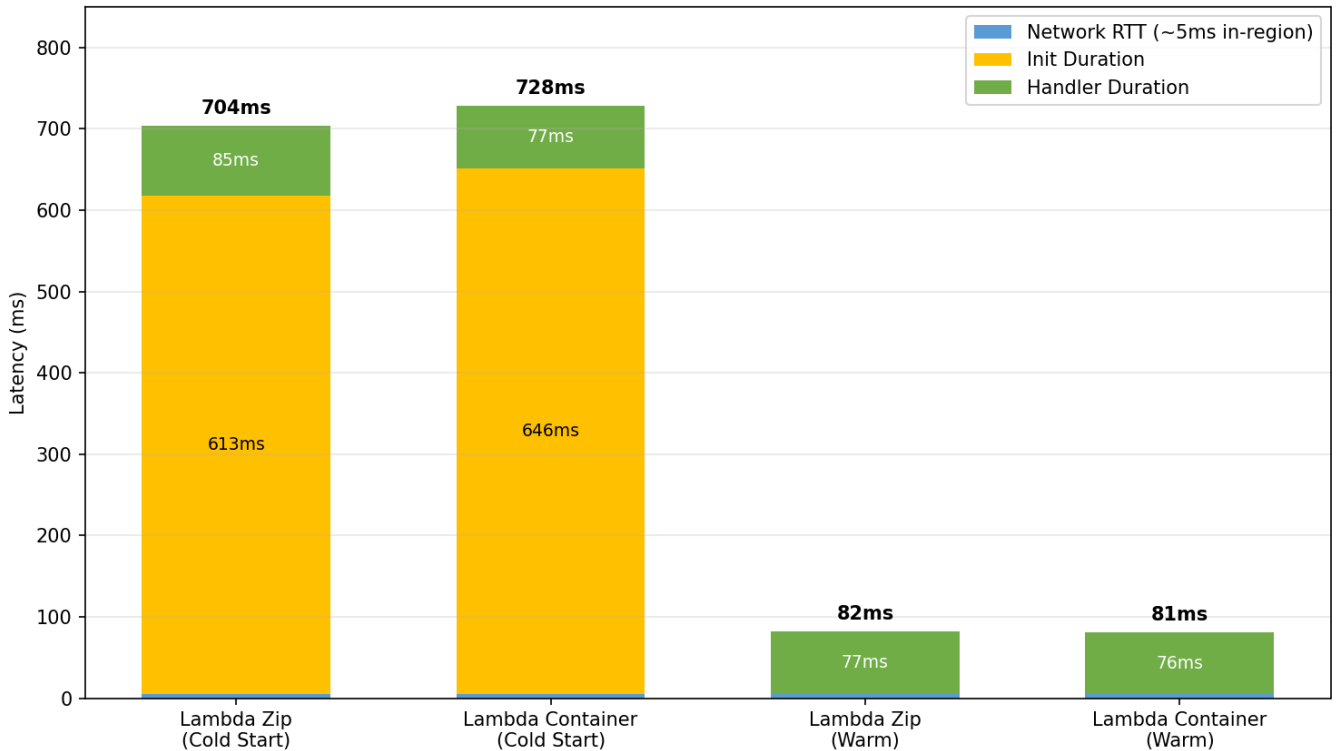
All endpoints successfully deployed. Verification confirmed identical k-NN results:

- **Fargate/ALB:** Index 35859, Distance 12.001459121704102, Query Time 22.43 ms.
- **EC2 Instance:** Index 35859, Distance 12.001459121704102, Query Time 33.598 ms.

2. Scenario A: Cold Start Characterization

Cold starts are triggered when no warm execution environment is available. For Lambda, this includes **Init Duration** (microVM and runtime setup) and **Handler Duration**.

Figure 1: Lambda Latency Decomposition — Cold Start vs Warm



Analysis:

- **Zip vs. Container:** Lambda Zip cold starts (704ms) are slightly faster than Container cold starts (728ms) due to lower image pull/extraction overhead.
- **Warm Baseline:** Warm invocations drop to ~81-82ms total, as the ~600ms Init Duration is eliminated.

3. Scenario B: Warm Steady-State Throughput

Steady-state testing measures how environments handle sustained load and concurrency.

Figure 4: Warm Steady-State Latency (Scenario B)
* Fargate c=50 shows queuing due to single-task capacity limit

Environment	Concurrency	p50 (ms)	p95 (ms)	p99 (ms)	Server avg (ms)
Lambda (zip)	10	69	82	87	65.2
Lambda (zip)	50	70	85	93	66.0
Lambda (container)	10	69	81	87	64.6
Lambda (container)	50	69	83	92	64.3
Fargate	10	28	31	35*	23.2
Fargate	50	142	287	435*	23.2
EC2	10	27	33	38	22.6
EC2	50	128	203	268	22.6

Analysis:

- **Scaling:** Lambda maintains stable latency at higher concurrency because it scales horizontally per request.

- **Queuing:** Fargate and EC2 show significantly higher p50/p99 at c=50 (142ms/435ms and 128ms/268ms) due to request queuing on a single-task capacity limit.

4. Scenario C: Burst from Zero

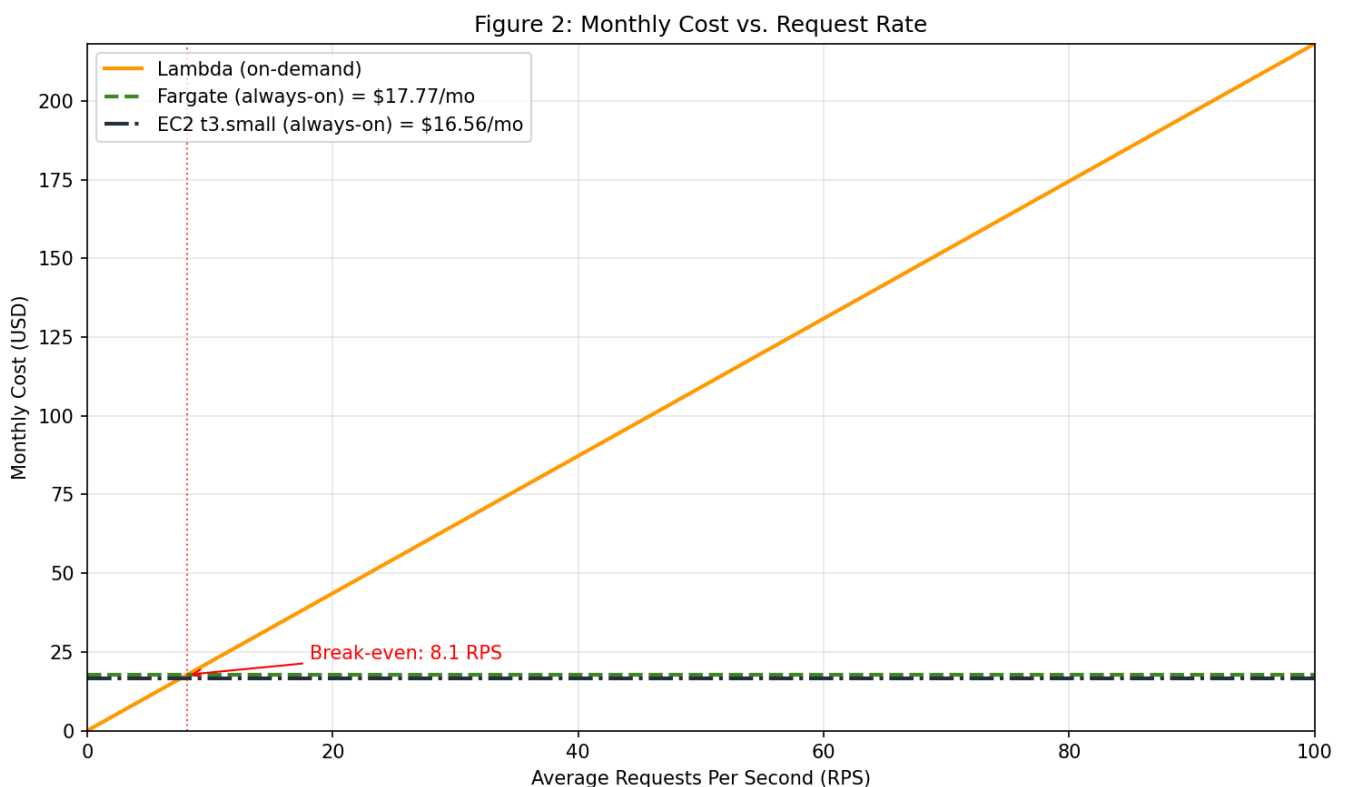
Simulating a traffic spike after 20 minutes of inactivity highlights the impact of cold starts on tail latency.

Analysis:

- **SLO Violation:** Under burst conditions, Lambda (Zip and Container) fails the p99 < 500ms SLO because the first requests trigger cold starts (p99 > 700ms).
- **Container Resilience:** EC2 and Fargate remain under 200ms p99 because the services are always warm and do not suffer from per-request provisioning delays.

5. Cost Analysis

Monthly costs vary between Lambda's per-request model and the "always-on" models of Fargate and EC2.

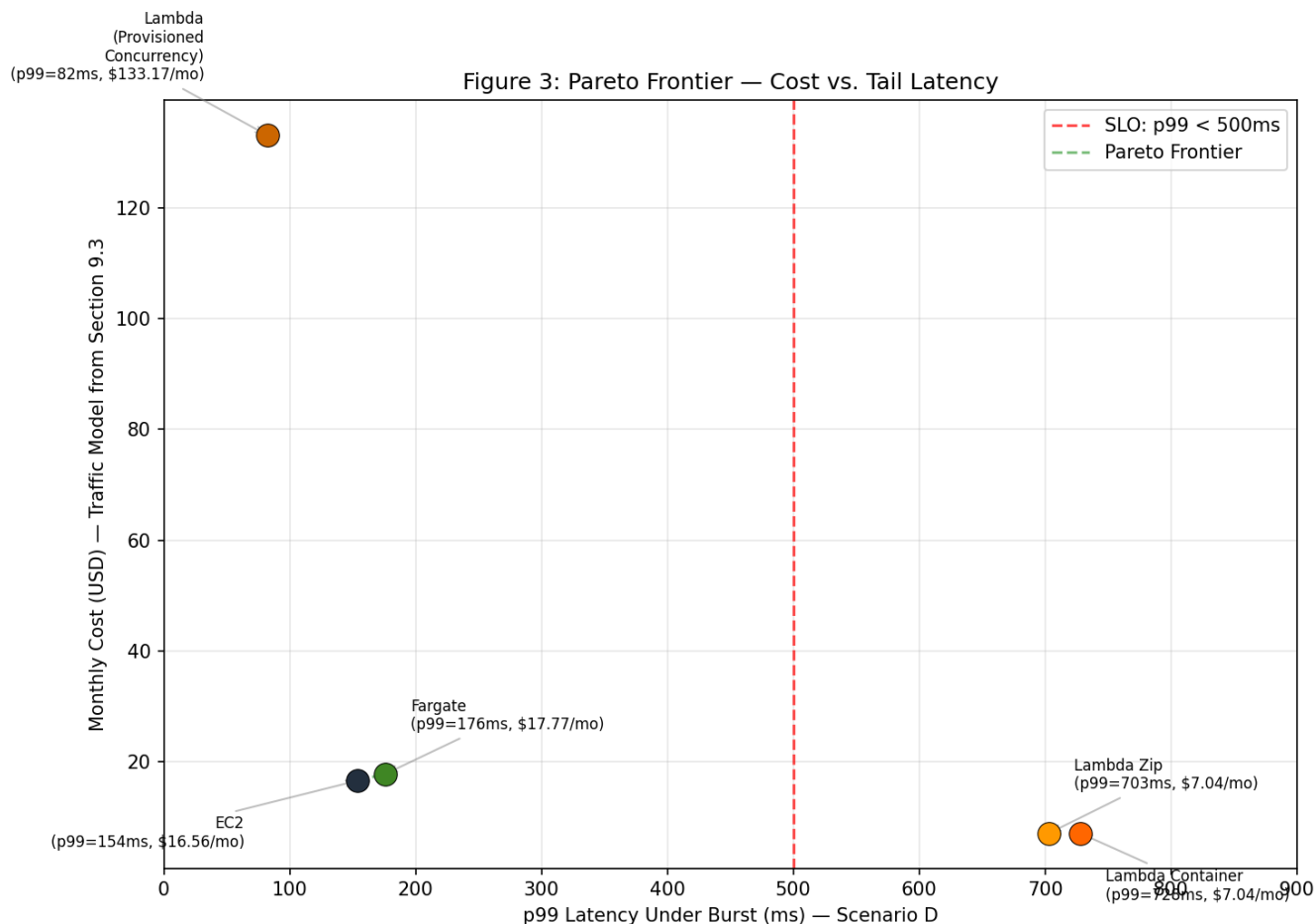


Idle Cost (Monthly):

- **Lambda:** \$0.00 (Zero idle cost).
- **Fargate:** \$17.77 (vCPU + RAM per hour).
- **EC2:** \$16.56 (t3.small instance per hour).

Break-Even Point: Lambda remains the most cost-effective choice for low-traffic scenarios, with a break-even point at **8.1 average RPS**.

6. Final Recommendation



Recommendation: EC2 t3.small (or Fargate if serverless management is preferred).

- **Justification:** While Lambda is cheaper at low volumes, it fails the p99 < 500ms SLO during bursts due to cold starts. EC2 and Fargate provide the best balance of cost and performance (Pareto-optimal) while meeting the SLO.
- **Alternative:** To meet the SLO with Lambda, **Provisioned Concurrency** could be used, but this increases the monthly cost to ~\$133.17, making it significantly more expensive than containerized options.